

Project: Temperature and Humidity Monitoring System with data logging

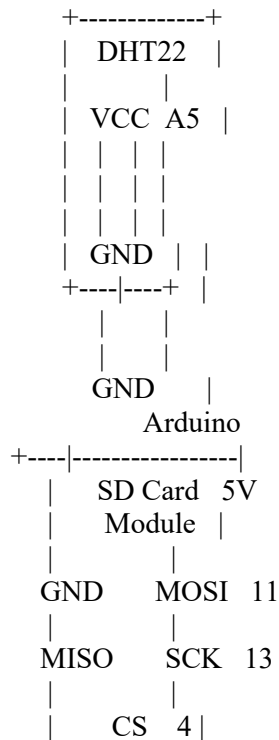
Introduction

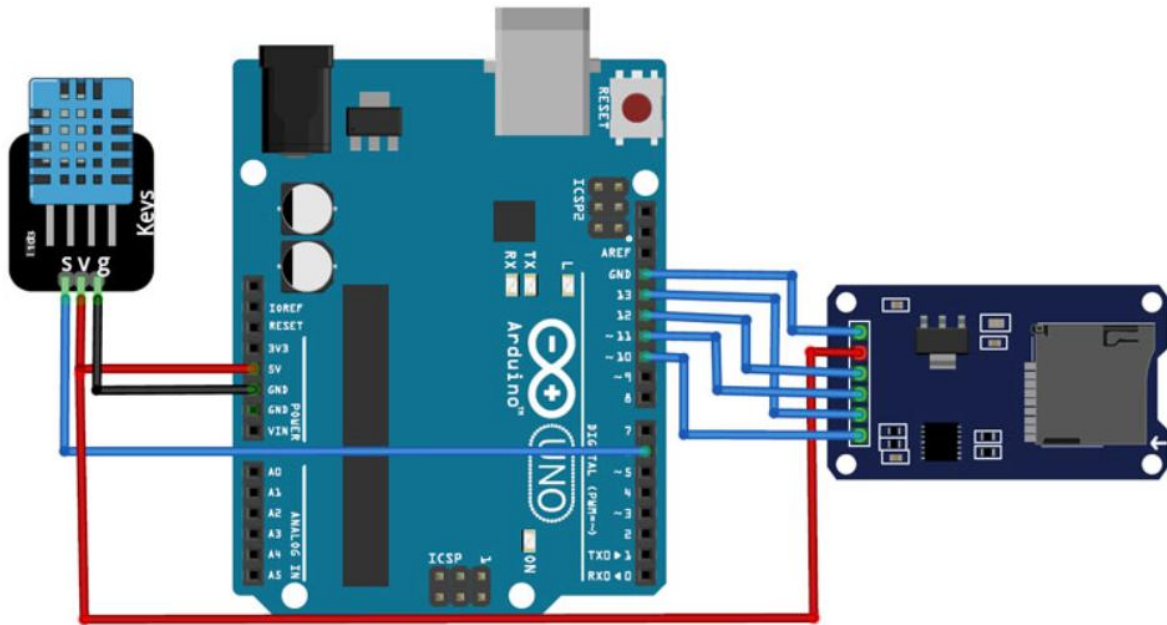
The aim of this project is to design and implement a temperature and humidity monitoring system using a DHT sensor and an SD card for data logging. This system records environmental data at regular intervals and stores it for future analysis.

Components and Tools

- Arduino Board
- DHT Sensor (DHT22)
- SD Card Module
- Jumper Wires
- Breadboard
- Software: Arduino IDE

Circuit Diagram





Code

```

cpp
#include <SPI.h>
#include <SD.h>
#include "DH".h"

#define DHTPIN A5
#define DHTTYPE DHT22 // Change to DHT22 if you're using that sensor
#define CSPIN 4 // Chip select pin for the SD card

DHT dht(DHTPIN, DHTTYPE);
File myFile;

void setup() {
  Serial.begin(9600);
  dht.begin();
  Serial.print("In`tializing SD card..."); if (!SD.begin(CSPIN)) {
    Serial.println("In`tialization failed!"); return;
  }
  Serial.println("In`tialization done.");}

void loop() {
  float humidity = dht.readHumidity();
  float temperatureC = dht.readTemperature();
  float temperatureF = dht.readTemperature(true);

  if (isnan(humidity) || isnan(temperatureC) || isnan(temperatureF)) {
    Serial.println("Fa`led to read from DHT sensor!");
    return;
  }

  long timestamp = millis();

  myFile = SD.open("dataLog.txt", FILE_WRITE);
  if (myFile) {

```

```

    myFile.print("Humidity: ");
    myFile.print(humidity);
    myFile.print("%, Temperature: ");
    myFile.print(temperatureF);
    myFile.print("°F, ");
    myFile.print(temperatureC);
    myFile.print("°C, Timestamp: ");
    myFile.println(timestamp);
    myFile.close();
    Serial.println("Data written to SD card.");
} else {
    Serial.println("Error opening file!");
}

Serial.print("Humidity: ");
Serial.print(humidity);
Serial.print("%, Temperature: ");
Serial.print(temperatureF);
Serial.print("°F, ");
Serial.print(temperatureC);
Serial.print("°C, Timestamp: ");
Serial.println(timestamp);

delay(10000); // Wait for 10 seconds before the next reading
}

```

Working Principle

The DHT sensor reads the temperature and humidity from the environment and sends this data to the Arduino. The Arduino then stores this data on an SD card along with a timestamp. The stored data can be retrieved later for analysis.

Results:

The system successfully logs temperature and humidity data to the SD card at regular intervals. The data can be analyzed to understand environmental conditions over time.

Log file Output:

Conclusion

This project demonstrates a simple yet effective way to monitor and log environmental data using an Arduino, DHT sensor, and SD card module. It can be further expanded by adding features like real-time data display on an LCD screen or remote monitoring through IoT.

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